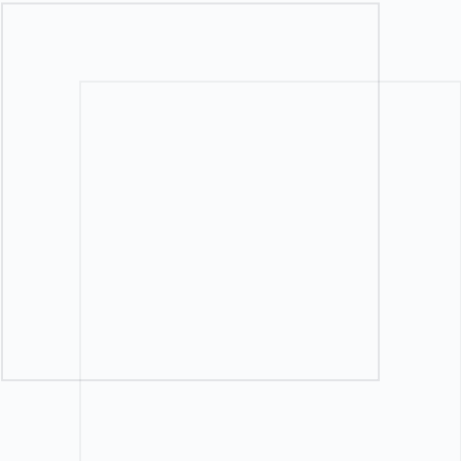




IMPLEMENTATION PLAN V1.2

BOS Light

*Data Contracts, Persistence Matrix, Acceptance Tests
и Cutline для разработки*

Версия 1.2 — финальная перед стартом
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1. Позиционирование

Это не BOS Kernel. Не BOS Runtime. Не Event Sourcing. Не Policy Engine.

BOS Light — organizational intelligence layer, который Paperclip исполняет. Paperclip = system of record и execution plane (agents, issues, budget, heartbeat, governance, events, audit). BOS = overlay: кто какие решения принимает, как задачи классифицируются, как снизить когнитивную нагрузку.

Три слоя

Concept Layer (BOS): 7 divisions, roles, VFP, BPI, Betting Table, Cynefin/OODA, Circuit Breaker, Eval Gates, guardrails.

Adaptation Layer (Plugin): BOS-метаданные как overlay (bpi_score, bos_status, blueprint_id, gate results) — stored in plugin state / Paperclip-native artifacts.

Execution Layer (Paperclip): Agents, issues, status, budget, heartbeat, events, UI, database — ground truth, не меняется.

2. Umbrella Risk: Plugin Runtime Caveat

PLUGIN_SPEC — proposed complete spec, не implementation contract

Paperclip plugin spec описывает богатую модель (tools, events, state, widgets, capability gates), но явно предупреждает: *текущая plugin runtime / admin UI — ранняя реализация, suitable for local/self-hosted persistent deployments, не для cloud/multi-instance distribution.*

Impact: capabilities, event emission, state API могут меняться. Plugin-first подход несёт architectural risk выше #3789 и #4093.

Mitigation: template-first (Phase 1) не зависит от plugin runtime. Plugin (Phase 2+) — thin adapter, легко переписывается при breaking changes. Не хранить durable organizational truth только в private plugin state.

3. MVP Definition of Done

BOS Light MVP считается готовым, если:

1. Импортируется company template с 7 division agents;
2. Каждый agent имеет AGENTS.md / hat-profile / VFP / guardrails;
3. Issue получает bpi_score и producer_division;
4. piko:blueprint-gen создаёт 5-секционный issue document;
5. Betting Table показывает batch из candidate issues;
6. Approve action не принимает approval itself, а создаёт Paperclip-native approval/request;
7. Хотя бы 3 Eval Gates пишут результат в issue/comment/document;
8. Circuit Breaker работает через polling fallback, если events не приходят;
9. Все BOS-данные имеют выбранное место хранения и migration path.

4. Data Contracts

4.1 BPI Score Schema

```
interface BPIScore {
  schema_version: "1.0";
  score: number; // 0.0 to 1.0, calculated
  formula: "bpi_v1.0";
  // Deterministic formula:
  //   normalized_cost = clamp01(estimated_token_cost / company_token_budget_ref)
  //   raw_score = (expected_value * urgency) / max(0.01, normalized_cost * risk_factor)
  //   score = clamp01(raw_score)
  components: {
    expected_value: number; // 0.0 to 1.0
    urgency: number; // 0.0 to 1.0
    estimated_token_cost: number;
    risk_factor: number; // 1.0 = baseline, >1.0 increases cost
    company_token_budget_ref: number; // token budget for normalization
  };
  hard_gates: {
    strategic_weight_passed: boolean;
    budget_snapshot_available: boolean;
    acceptance_inputs_present: boolean;
    policy_precheck_passed: boolean; // P0: lightweight, not full LARS
    security_precheck_passed: boolean; // P0: basic scan, not full Eval Gate
  };
  scored_by: "Div2.MasterPlanner";
  scored_at: string; // ISO 8601
  source_issue_id: string;
}
```

4.2 BOS Status Overlay

```
interface BosStatusOverlay {
  schema_version: "1.0";
  issue_id: string; // Paperclip issue UUID
  bos_status: BosStatus; // enum below
  producer_division: Division;
  blueprint_id: string | null;
  bpi_score: number | null;
  updated_at: string;
}

enum BosStatus {
  RAW_ORDER = "RAW_ORDER",
  TRIAGED = "TRIAGED",
  SHAPING = "SHAPING",
  BLUEPRINT_READY = "BLUEPRINT_READY",
  BETTING_POOL = "BETTING_POOL",
  APPROVAL_REQUESTED = "APPROVAL_REQUESTED",
  APPROVED_FOR_CYCLE = "APPROVED_FOR_CYCLE",
  IN_PROGRESS = "IN_PROGRESS",
  QA_REVIEW = "QA_REVIEW",
  CORRECTION_REQUIRED = "CORRECTION_REQUIRED",
  ACCEPTED = "ACCEPTED",
  RELEASED = "RELEASED",
  ARCHIVED = "ARCHIVED",
  REJECTED_BPI = "REJECTED_BPI",
  CIRCUIT_BREAKER_OPEN = "CIRCUIT_BREAKER_OPEN",
  FAILED = "FAILED"
}
```

4.3 Betting Table Item

```
interface BettingTableItem {
  schema_version: "1.0";
  cycle_id: string;           // "cycle_2026w22"
  issue_id: string;
  bpi_score: number;
  blueprint_id: string | null;
  status:
    | "CANDIDATE"
    | "APPROVAL_REQUESTED"      // Approve button clicked, native request created
    | "APPROVED_FOR_CYCLE"      // Native approval confirmed by governance
    | "REJECTED"
    | "COMPLETED";
  native_approval_request_id: string | null; // Paperclip-native approval ID
  native_approval_status: "PENDING" | "APPROVED" | "REJECTED" | null;
  approved_by: string | null; // agent_id or "Master.Human"
  created_at: string;
  updated_at: string;
}
```

4.4 Eval Gate Result

```
interface EvalGateResult {
  schema_version: "1.0";
  issue_id: string;
  run_id: string | null;
  gates: Array<{
    gate_id: string;           // "LARS.Deterministic" | "LARS.SecurityPolicy" | "LARS.ArtifactIntegrity" |
    "LARS.Budget"
    status: "PASSED" | "FAILED" | "NOT_RUN";
    evidence: string | null;
    is_blocking: boolean;
  }>;
  overall: "PASSED" | "PASSED_WITH_WARNINGS" | "FAILED_BLOCKING" | "INCOMPLETE";
  blocking_failure_count: number;
  warning_count: number;
  not_run_count: number;
  evaluated_at: string;
  evaluated_by: "Div5.Qualifications";
}

const P1_GATES = [
  { id: "LARS.Deterministic", blocking: true, criteria: "Required fields present, schema valid" },
  { id: "LARS.SecurityPolicy", blocking: true, criteria: "No direct access/mutation, tool scope respected" },
  { id: "LARS.ArtifactIntegrity", blocking: true, criteria: "Output matches blueprint, no hallucinated fields" },
  { id: "LARS.Budget", blocking: false, criteria: "Spend within grant, alert at 80%" }
];
```

4.5 Circuit Breaker Record

```
interface CircuitBreakerRecord {
  schema_version: "1.0";
  issue_id: string;
  state: "CLOSED" | "HALF_OPEN" | "OPEN";
  attempt_count: number; // increments on each failed run
  max_attempts: number; // default: 3
  half_open_threshold: number; // default: 2
  last_failure_at: string | null;
  last_failure_reason: string | null;
  opened_at: string | null;
  escalation_issue_id: string | null; // created when OPEN
  updated_at: string;
}

// Fallback: polling – active runs only, with jitter and backoff
const POLLING_CONFIG = {
  poll_scope: "ACTIVE_RUNS_ONLY", // do not scan archived/completed issues
  interval_ms: 30000,
  jitter_ms: 5000, // randomize to avoid thundering herd
  backoff_after_attempts: 10, // slow down after 10 empty polls
  max_retries: 60, // 30 min timeout
  fallback_source: "activity_log" // if events unavailable
};
```

4.6 Decision Metadata

```
interface DecisionMetadata {
  schema_version: "1.0";
  decision_id: string;
  issue_id: string;
  cynefin_domain: "CLEAR" | "COMPLICATED" | "COMPLEX" | "CHAOTIC" | "DISORDER";
  confidence: number; // 0.0 to 1.0
  decision_type: "POLICY_UPDATE" | "BATCH_APPROVAL" | "RUSH_AUTHORIZATION" | "EXPERIMENT" | "SELF_HEALING";
  emitted_events: string[];
  recommended_action: string;
  decided_by: "Div7.Executive";
  decided_at: string;
}
```

5. Persistence Strategy Matrix

Таблица 1 Где хранить BOS-данные

BOS Object	Paperclip Surface	Primary Storage	Fallback	Acceptance Test
BPI Score	Issue annotation / plugin data tab	Issue-scoped plugin state	Issue Document / comment	Score visible after issue.created
BOS Status	Issue detail tab overlay	Issue-scoped plugin state	Issue label / comment	bos_status synced with issue lifecycle
Blueprint	Issue Document (5 sections)	Issue Document (native)	Issue description	Blueprint generated on approval
Betting Table	Dashboard widget	Plugin entity / config / current cycle	Managed Paperclip issue/project	Approve creates native approval request
Gate Result	Review checklist / comment	Issue-scoped state + Issue Document	Issue comment	Blocking gate stops release path
Circuit State	Dashboard / detail tab	Issue-scoped plugin state	Polling + activity log scanning	3 failures → OPEN + escalation issue
BOS Config	Plugin config	Config JSON	Company template import files	Reload without data loss
Decision Record	Issue comment / document	Paperclip issue comment (native)	Issue Document	Decision visible in issue history

Правило: durable truth в Paperclip-native artifacts

Важные решения (Betting Table approval, gate results, decision records) отражаются в Paperclip-native artifacts: issue comments, issue documents, approvals, activity log. BOS overlay не становится скрытым вторым источником истины.

6. Acceptance Tests

Таблица 2 Acceptance Tests по P0/P1 компонентам

#	Компонент	Given	When	Then
A1	Company Template	Fresh Paperclip company	Import BOS template (companies.sh)	7 agents created with AGENTS.md, org chart rendered
A2	BPI Score	Issue created in backlog	Div2 agent calls piko:bpi-score	Score 0.0-1.0 stored in issue-scoped state, visible in UI
A3	Blueprint Gen	Issue approved by BPI	piko:blueprint-gen invoked	Issue Document with 5 sections (identity, bpi, acceptance, resources, qa)
A4	Betting Table	5+ issues with BPI score	Widget renders Pitch Deck	Table shows top-N by BPI, Approve button visible
A5	Batch Approval	Betting Table with candidates	User clicks Approve Batch	Paperclip-native approval/request created, no plugin-side decision
A6	Eval Gate: Pass	Issue in QA_REVIEW	All blocking gates pass	Status ACCEPTED, gate results in issue document
A7	Eval Gate: Fail	Issue in QA_REVIEW	Blocking gate fails	Status CORRECTION_REQUIRED, guidance attached
A8	Circuit Breaker: Close	Issue with 1 failed attempt	Retry succeeds	State CLOSED, attempt_count reset
A9	Circuit Breaker: Open	Issue with 3 failed attempts	3rd failure detected	State OPEN, escalation issue created, alert sent
A10	CB Fallback	Events unavailable (#3789)	Circuit Breaker active	Polling detects failures, transitions correct
A11a	State: Config	Plugin state cleared	Restart plugin	BOS config reloaded from Paperclip artifacts
A11b	State: BPI Scores	Plugin state cleared	Restart plugin	BPI scores restored from issue documents/state
A11c	State: Betting Table	Plugin state cleared	Restart plugin	Current cycle restored from native issue/project
A11d	State: Gate Results	Plugin state cleared	Restart plugin	Gate results restored from issue comments/docs
A11e	State: Decisions	Plugin state cleared	Restart plugin	Decision records restored from issue comments

7. План фаз (реалистичный)

Временная шкала: 3 недели = demonstrable vertical slice

5-6 недель = надёжный MVP с тестами, fallback persistence, нормальным UX. Не production-ready, но runnable and testable.

Phase 0: Positioning (2 дня)

- Discord #dev: pitch BOS Light как company template + plugin
- Не kernel, не runtime, не policy engine
- Результат: conceptual approval

Phase 1: Company Template (1 неделя) — не зависит от plugin

- 7 AGENTS.md с hat-profiles, VFP, guardrails
- Org chart, reporting lines, task routing
- Ритуалы: daily pulse, weekly review, batch approval ritual
- Acceptance: A1 passes

Phase 2: State Spike + Minimal Plugin (1-2 недели)

- **Day 1: State Spike** — проверить ctx.state per-company (#4093), issue-scoped state, fallback config JSON
- **piko:bpi-score** — agent tool, score → issue-scoped state (A2)
- **piko:blueprint-gen** — Issue Document template, 5 sections (A3)
- **Betting Table widget** — dashboard, Pitch Deck, Approve → native approval (A4, A5)

Phase 3: Safety Loop (1 неделя)

- Circuit Breaker: polling-based (fallback от #3789), 3 states (A8, A9, A10)
- Eval Gates: 4 gates (Deterministic, SecurityPolicy, ArtifactIntegrity, Budget) (A6, A7)
- Gate results → issue document + comment

Phase 4: Div7 Decision Protocol (1-2 недели) — после usage traces

- piko:decide — Cynefin classification + OODA loop
- Обучение на реальных traces: incident types, budget escalations, strategic decisions
- Decision results → Paperclip issue comments (native artifacts)

8. Demo Script

5-минутная демонстрация BOS Light

1. **Import (30s):** Import BOS company template → 7 agents appear, org chart visible
2. **Create Issue (30s):** User creates vague issue → Div2 triggers piko:bpi-score → score appears in issue tab
3. **Blueprint (30s):** Approve issue → piko:blueprint-gen → Issue Document with acceptance criteria
4. **Betting Table (60s):** Create 4-5 more issues → Dashboard shows Pitch Deck sorted by BPI → Click Approve Batch → Paperclip approval request created
5. **Safety Loop (90s):** Agent completes work → Eval Gates run → Gate results in issue document → Circuit Breaker visible in dashboard
6. **Failure (60s):** Trigger 3 failures → Circuit Breaker goes OPEN → Escalation issue auto-created

9. Чеклист перед стартом

Таблица 3 Предусловия разработки

#	Проверка	Метод	Статус
C1	Conceptual approval в Discord #dev	Pitch + feedback thread	TODO
C2	Event emission #3789 — проверить	Plugin test: subscribe + trigger failure	TODO
C3	Plugin state #4093 — проверить	State spike: write company-scoped, read back	TODO
C4	Company template import/export	Test companies.sh с 7 agents	TODO
C5	AGENTS.md syntax compatibility	Validate against Paperclip agent config schema	TODO
C6	Plugin runtime version check	Confirm min version, check for breaking changes	TODO
C7	Plugin capability set confirmed	Verify all needed caps available in current runtime	TODO

Финальная формула

BOS Chimera 4.1 не портируется в Paperclip как kernel. BOS редуцируется до organizational intelligence layer: company template, role semantics, prioritization, batch governance, decision protocols, quality gates. Paperclip остаётся system of record и execution plane. Все BOS-состояния, grants, decisions и gates — overlay metadata, а не replacement core logic. 3 недели — demonstrable vertical slice. 5-6 недель — runnable MVP.